NLP 053 - CSE UOI 2026
Kaggle Challenge: Food Hazard Detection

Food Hazard Prediction from Recall Texts (SemEval-2025 Task 9, ST1)

The Food Hazard Detection task evaluates explainable classification systems for titles of food-incident reports collected from the web.

Kaggle link: https://www.kaggle.com/t/e6a57812ae554144a90823a7ddf48fd0

Official Semeval link: https://food-hazard-detection-semeval-2025.github.io/

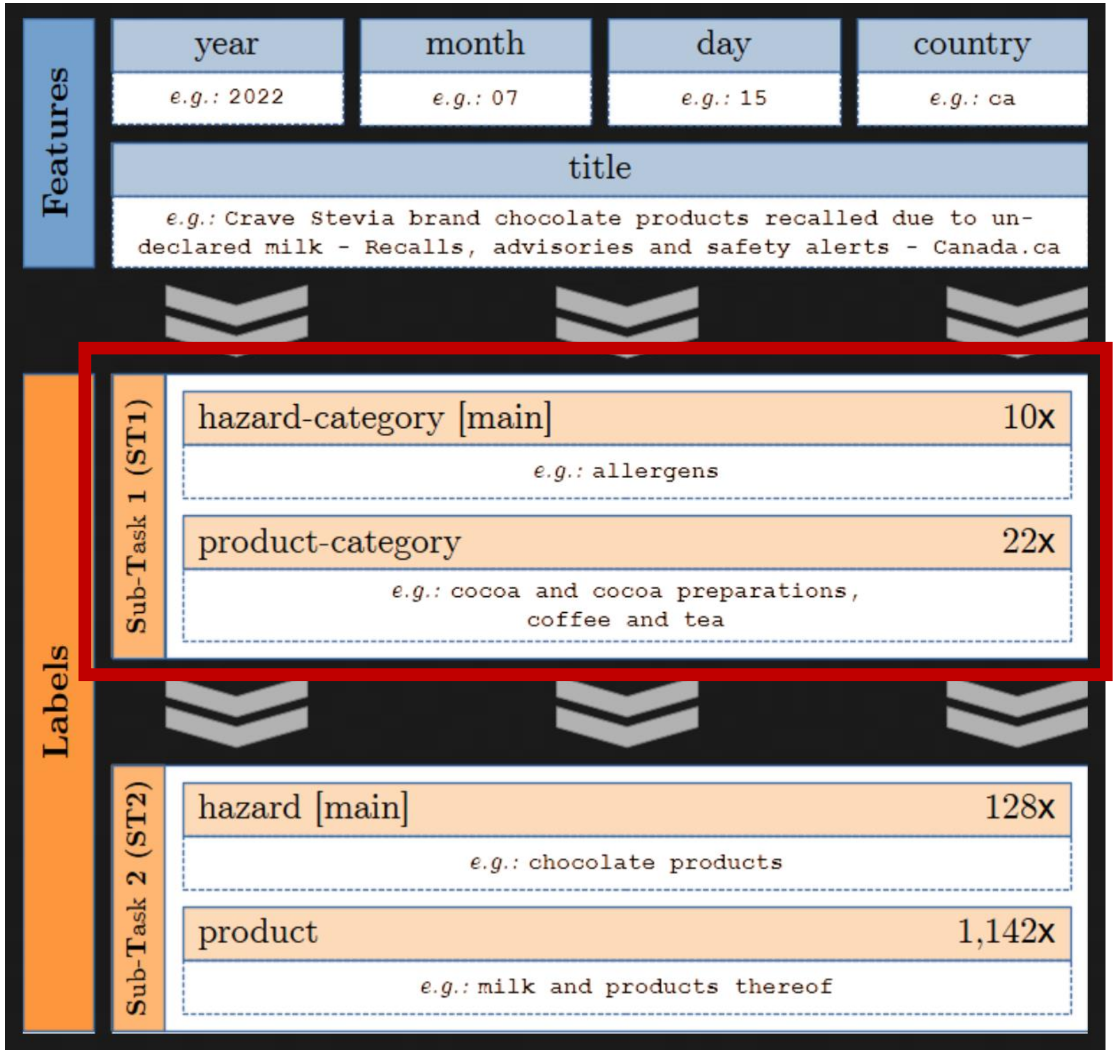
SemEval-2025 Task 9, Subtask 1 (ST1)

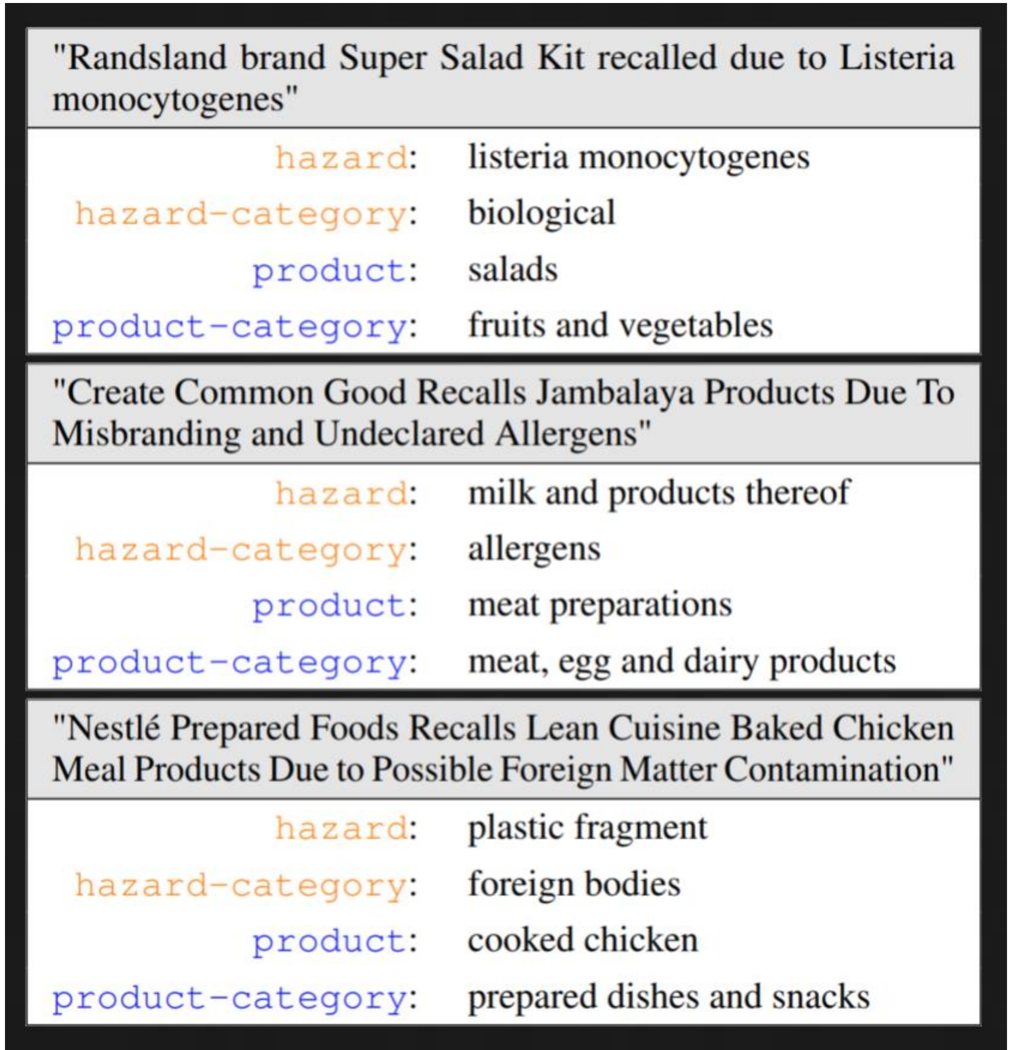
- **Food hazard-category and product-category classification**. In ST1, each instance is a food recall report, and the model must predict two coarse-grained labels from the text: a **hazard-category** with **10 classes** and a **product-category** with **22 classes**.
- The official challenge used mostly short English recall titles, though the dataset also contains full recall text and metadata such as date and country. The task is intentionally difficult because the label distribution is **long-tail / heavily imbalanced**.
- The official score is not just “average of two classifiers.” It is: **(macro-F1 for hazard-category + macro-F1 for product-category on only the examples where hazard-category was predicted correctly) / 2**. That means the task strongly prioritizes getting the hazard right; if hazard is wrong, product prediction does not help on that example.
- **Think about class imbalance, macro-F1, and error propagation!**

Stats

- The official SemEval split was **5,082 train / 565 validation / 997 test**, for a total of **6,644 English instances** in the challenge.
- The organizers also released the broader **Food Recall Incidents** dataset on Zenodo under **CC BY-NC-SA 4.0**; that full release contains **7,546 texts across 6 languages**, with English being the dominant language, and includes a semeval-split field indicating which examples were used for the challenge.

Sub-task 1 and data

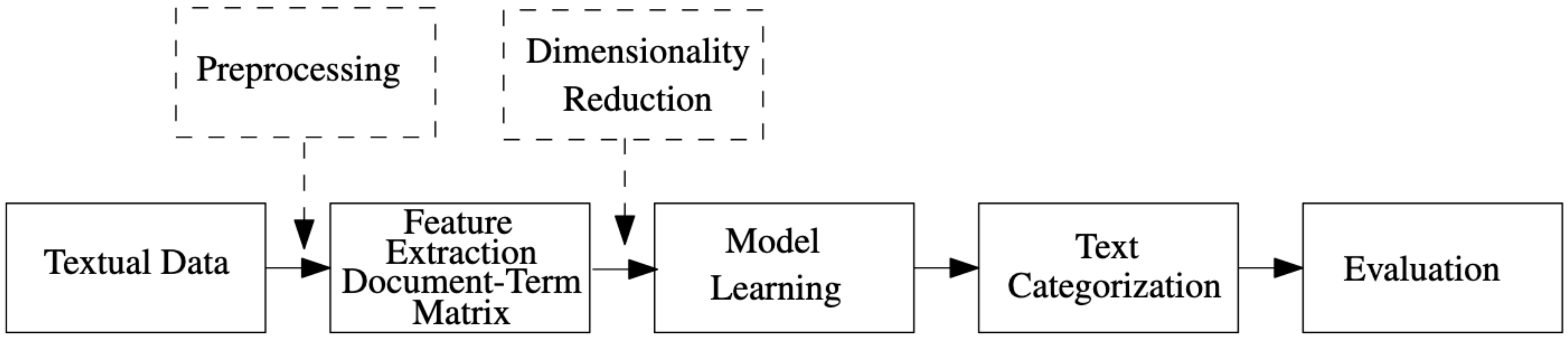




Key points

- Deadline: 01-06-2026 (3 μήνες)
- Ομάδες: 1-2 άτομα
- 5 submissions per day
- Python, Ipython Notebook, Jupyter (+ ότι βιβλιοθήκη θέλετε)
- Report & presentation
  - 1 άτομο->min 5σελιδο, 2 άτομα-> min 10σελιδο (best->Latex)
  - Presentation: 15 slides max
- **<u>Πρέπει να περάσετε τις προφορικές ή γραπτές εξετάσεις για να μετρήσει η εργασία σας!</u>**

What to show



- Να περιέχει ανάλυση δεδομένων
- Πώς αντιμετωπίσατε κάθε στάδιο του προβλήματος:
  - τι είδους αναπαράσταση δεδομένων χρησιμοποιήσατε, ποια χαρακτηριστικά χρησιμοποιήσατε,
  - αν εφαρμόσατε τεχνικές μείωσης των διαστάσεων
  - ποιους αλγορίθμους δοκιμάσατε και γιατί, την απόδοση των μεθόδων σας, χρόνος εκπαίδευσης
  - οποιεσδήποτε προσεγγίσεις που τελικά δεν λειτούργησαν αλλά είναι ενδιαφέρον να παρουσιάσετε, και γενικά, ό,τι νομίζετε ότι είναι ενδιαφέρον να αναφέρετε

Grading

- **15% Task understanding, setup, analysis**
  Correct use of data, labels, show preprocessing, stats.
- **10% Reproducible classical baselines**
  Clean implementation and reporting.
- **20% Neural baselines**
  Sensible neural approach and comparison against baseline.
- **20% Methodological improvement (e.g. SOTA)**
  A real idea, not random hyperparameter fishing.
- **15% Evaluation quality**
  Proper metric use, ablations, confusion analysis, class-wise discussion.
- **10% Final report**
  Clear writing, figures, limitations, related work.
- **10% Presentation / demo**
  Clear explanation, lessons learned, answering questions.

Notes

- Multi-class text classification (text categorization)
- Python packages: pandas, networkx, scikit-learn, spacy, nltk, genism
- Analysis, visualization, feature selection
- Cross-validation to get score locally (not on Kaggle)
- Word2vec, Glove, Fasttext, BERT embeddings (for text)
- Multiple classifiers
- SOTA methods (after you get a good score)

Extra points

- Most creative team name
  - Computer science + news
- Fastest solutions
  - Before I upload the first baseline
- Running SOTA method
  - Be the only one who used an approach

Have fun!